

9 (**long**) A rigid uniform plank of length $\sqrt{3}\ell$ rests on two supports with its midpoint located a distance $\ell/2$ from each support, as shown in Fig. 6(a). Support B fails suddenly and the plank rotates about support A before hitting a fixed horizontal rail C located a distance $\ell/2 + c\ell/2$ from A at angle θ_0 , as shown in Fig. 6(b).

(a) Assuming no slip at A, find an expression for the angular acceleration of the plank $\ddot{\theta}$ as a function of its rotation θ . [6]

(b) Find a similar expression for the angular velocity of the plank $\dot{\theta}$ and its value ω_0 at the moment of impact with the rail. [6]

(c) If the coefficient of restitution between the plank and the rail is e , calculate the subsequent angular velocity of the plank ω_1 assuming there is no impulsive reaction at support A. [8]

(d) By considering the subsequent velocity of point A on the plank, find the limiting value of the distance AC (i.e. the value of constant c) such that there is no impulsive reaction at support A. [6]

(e) Assuming an elastic impact with the rail and $c = 1$, find the greatest height reached by the midpoint of the plank during the subsequent motion. [4]

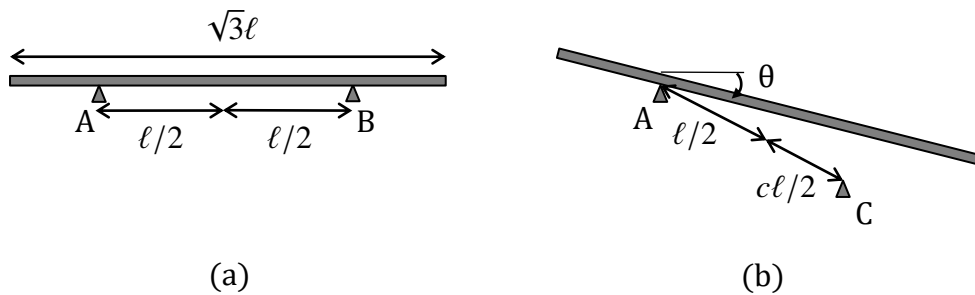


Fig. 6